



**NATIONAL
SENIOR CERTIFICATE/
NASIONALE
SENIOR SERTIFIKAAT**

GRADE/GRAAD 11

NOVEMBER 2022

**TECHNICAL SCIENCES P1/
TEGNIJSE WETENSKAPPE V1
MARKING GUIDELINE/NASIENRIGLYN**

MARKS/PUNTE: 150

This marking guideline consists of 9 pages./
Hierdie nasienriglyn bestaan uit 9 bladsye.

QUESTION/VRAAG 1

- | | | |
|------|------|-------------|
| 1.1 | B ✓✓ | (2) |
| 1.2 | B ✓✓ | (2) |
| 1.3 | D ✓✓ | (2) |
| 1.4 | B ✓✓ | (2) |
| 1.5 | A ✓✓ | (2) |
| 1.6 | D ✓✓ | (2) |
| 1.7 | C ✓✓ | (2) |
| 1.8 | A ✓✓ | (2) |
| 1.9 | B ✓✓ | (2) |
| 1.10 | C ✓✓ | (2) |
| | | [20] |

QUESTION/VRAAG 2

- | | | |
|-----|-----|------------|
| 2.1 | B ✓ | (1) |
| 2.2 | D ✓ | (1) |
| 2.3 | A ✓ | (1) |
| 2.4 | C ✓ | (1) |
| 2.5 | F ✓ | (1) |
| 2.6 | E ✓ | (1) |
| 2.7 | H ✓ | (1) |
| 2.8 | I ✓ | (1) |
| | | [8] |

QUESTION/VRAAG 3

3.1 Experiment 1/Eksperiment 1 ✓ (1)

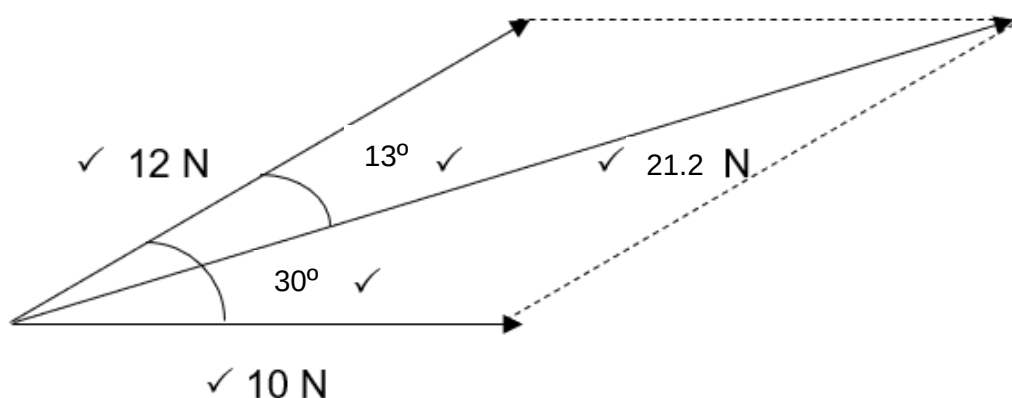
3.2 Resultant = $12 + (-10) = 2 \text{ N}$ ✓✓ (2)

3.3 $R^2 = (12)^2 + (10)^2$ ✓
 $R = 15,62 \text{ N}$ ✓
 $\tan \theta = \frac{\text{opp./teenoorst.}}{\text{adj./langsaaan}} = \frac{10}{12}$ ✓
 $\theta = 39,8^\circ$ ✓

$R = 15,62 \text{ N}$ at an angle $39,8^\circ$ with 12 N vector ✓

$= 15,62 \text{ N}$ teen 'n hoek van $39,8^\circ$ met die 12 N vector. (5)

3.4 Pythagoras theorem / Pythagoras se stelling ✓ (1)



3.5 Resultant = $21,2 \text{ N}$ at an angle 13° with 12 N ✓
 Resultant = $21,2 \text{ N}$ teen 'n hoek van 13° met 12 N ✓
 Accept/Aanvaar: ($21 \text{ N} - 21,5 \text{ N}$ and/en $13^\circ - 13,6^\circ$) (6)

3.6 The magnitude of the resultant increases with decrease in the angle between the forces. ✓✓/OR The magnitude of the resultant decreases with an increase in the angle between the forces.

Die grootte van die resultant neem toe met 'n afname in die hoek tussen die kragte. ✓✓/ OF Die grootte van die resultant neem af met 'n toename in die hoek tussen die kragte.

(2)
[17]

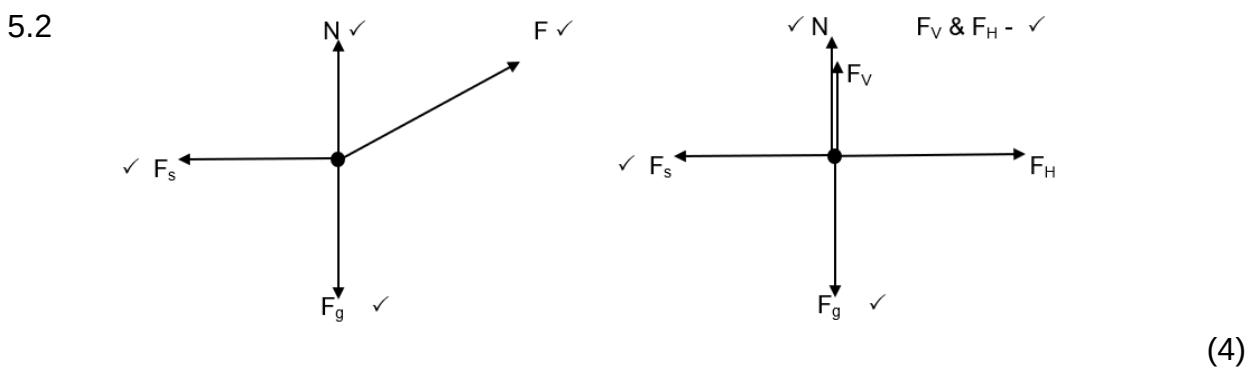
QUESTION/VRAAG 4

4.1 $F_{1(\text{hor})} = F_1 \cos \theta$ ✓
 $F_{1(\text{hor})} = (150)(\cos 60^\circ)$
 $F_{1(\text{hor})} = 75 \text{ N}$ ✓
 $F_{1(\text{hor})} = F_{2(\text{hor})}$ (horizontal forces are balanced)
(horizontale kragte is gebalanseerd) ✓
 $75 = F_2 \cos 68^\circ$
 $F_2 = 200,21 \text{ N}$ ✓ (4)

4.2 $F_{1(\text{ver})} = F_1 \sin \theta$ $F_{2(\text{ver})} = F_2 \sin \theta$ (for both formulae ✓)
(vir albei formules ✓)
 $F_{1(\text{ver})} = (150) \sin 60^\circ$ $F_{2(\text{ver})} = (200,21) \sin (68^\circ)$
 $= 129,9 \text{ N}$ ✓ $= 185,63 \text{ N}$ ✓
 $F_{\text{ver up}} (F_{\text{ver op}}) = 129,9 + 185,63 = 315,53 \text{ N}$ ✓
 $F_{\text{ver up}} = F_{\text{ver down}} (F_{\text{ver af}})$ ✓
 $315,53 = mg$
 $315,53 = (m)(9,8)$
 $m = 32,2 \text{ kg}$ ✓ (6)

[10]**QUESTION/VRAAG 5**

5.1 The force that opposes the tendency of motion of a stationary object relative to a surface. /The force acts between the two surfaces when the object is stationary. ✓✓
Die krag wat die neiging van beweging van 'n stilstaande voorwerp relatief tot 'n oppervlak teenwerk. / Die krag werk tussen die twee oppervlaktes wanneer die voorwerp in rus is. ✓✓ (2)



5.3 $f_s = \mu_s N$ ✓
 $f_s = (0,4) \left(\frac{1}{3} \right) (50)(9,8)$ ✓
 $f_s = 65,33 \text{ N}$ ✓ (3)

5.4 $f_s = F_H$ ✓
 $65,33 = F \cos 35^\circ$ ✓✓
 $F = 79,75 \text{ N}$ ✓ (4)

[13]

QUESTION/VRAAG 6

6.1 6.1.1 Longitudinal wave/Longitudinale golf ✓

The particles of medium vibrate parallel to the direction of the motion of the wave. ✓✓

Die deeltjies van die medium vibreer parallel aan die rigting van die beweging van die golf. ✓✓ (3)

6.1.2 Compression/Verdigting ✓ (1)

6.1.3 Rarefaction/Verdunning ✓ (1)

6.1.4 $\lambda = \frac{1,32}{2} \checkmark = 0,66 \text{ m} \checkmark$ (2)

6.1.5 $v = f \lambda \checkmark$
 $v = (512) (0,66) \checkmark$
 $v = 337,92 \text{ m.s}^{-1} \checkmark$ (3)

6.2 6.2.1 2 m ✓✓ (2)

6.2.2 2 s ✓✓ (2)

6.2.3 $v = f \lambda \checkmark$
 $10 = (\frac{1}{2}) \lambda \checkmark \checkmark$
 $20 \text{ m} = \lambda \checkmark$ (4)

[18]

QUESTION/VRAAG 7

- 7.1 7.1.1 D ✓
The speed of sound is the lowest in a gaseous substance. ✓✓
Die spoed van klank is die laagste in 'n gasagtige stof. ✓✓ (3)
- 7.1.2 A ✓
The speed of sound is the highest in a solid substance. ✓✓
Die spoed van klank is die hoogste in 'n vastestof. ✓✓ (3)
- 7.2 7.2.1 Echo/Eggo ✓ (1)
- 7.2.2 $v = f \lambda$ ✓
 $v = (32 \times 10^3) (0,05)$ ✓
 $v = 1600 \text{ m.s}^{-1}$ ✓
 $v = \frac{2d}{t}$
 $1600 = \frac{2d}{0,25}$ ✓
 $d = 200 \text{ m}$ ✓ (5)
- 7.3 7.3.1 Waveform/Golfvorm Q ✓
Amplitude of waveform P < Amplitude of waveform Q ✓
Higher the amplitude the louder the sound. ✓
Amplitude van golfvorm P < Amplitude van golfvorm Q ✓
Hoe hoër die amplitude, hoe harder is die klank. ✓ (3)
- 7.3.2 Pitch increases with frequency (Pitch is directly proportional to frequency) ✓✓
Toonhoogte neem met frekwensie toe (Toonhoogte is direk eweredig aan frekwensie.) ✓✓ (2)
- 7.4 7.4.1 Sound of frequencies between 20 kHz and 100 kHz ✓✓
Klank met frekwensies tussen 20 kHz en 100 kHz ✓✓ (2)
- 7.4.2 Ranges from 20 Hz to 20 kHz ✓✓
In 'n gebied van 20 Hz tot 20 kHz ✓✓ (2)

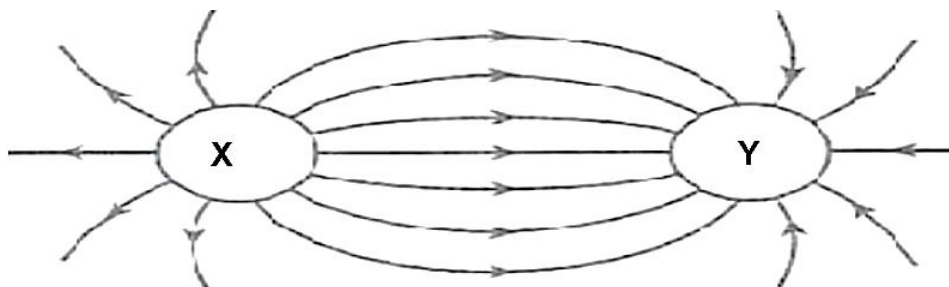
[21]

QUESTION/VRAAG 8

- 8.1 Attraction/Aantrekking ✓
 Unlike poles attract each other/Teenoorgestelde pole trek mekaar aan ✓✓ (3)
- 8.2 8.2.1 North pole/Noordpool ✓ (1)
- 8.2.2 **P** ✓
 Field lines are closer together at **P** ✓✓
 Veldlyne is nader aan mekaar by **P** ✓✓ (3)
- 8.3 Magnetic storm / Northern lights (Aurora Borealis) ✓
 Magnetiese storm / Noordelike ligte (Aurora Borealis) ✓ (1)

[8]**QUESTION/VRAAG 9**

9.1 9.1.1

**Marking guideline**

Correct shape ✓

Direction from positive to negative ✓

Field lines do not cross and start on the sphere ✓

Nasienriglyne

Korrekte vorm ✓

Rigting van positief na negatief ✓

Veldlyne kruis nie en begin by die sfeer. ✓

(3)

- 9.1.2 The magnitude of the electrostatic force exerted by one point charge on another point charge is directly proportional to the product of the (magnitude of the) charges ✓ and inversely proportional to the square of the distance between them. ✓
 Die grootte van die elektrostaties krag wat deur een puntlading op 'n ander puntlading uitoefen is direk eweredig aan die produk van die (groottes van die) ladings ✓ en omgekeerd eweredig aan die kwadraat van die afstand tussen hulle. ✓ (2)

$$9.1.3 \quad F = \frac{kQ_1Q_2}{r^2} \quad \checkmark$$

$$F = \frac{(9 \times 10^9)(5,56 \times 10^{-6})(2,56 \times 10^{-6})}{(0,01)^2} \quad \checkmark \checkmark$$

$$F = 1\,281,02 \text{ N} \quad \checkmark$$

(4)

9.1.4 ATTRACTION/AANTREKKEND

(1)

9.2 9.2.1 TOWARDS/NA Q ✓

Positive charge will move in the direction of the electric field.

OR Positive charge attracted towards negative plate. ✓✓*Positiewe lading sal beweeg in die rigting van die elektriese veld. **OF****Die positiewe ladings word aangetrek na die negatiewe plaat. ✓✓*

(3)

$$9.2.2 \quad E = \frac{V}{d} \quad \checkmark$$

$$E = \frac{200}{0,02} \quad \checkmark$$

$$E = 10\,000 \text{ N.C}^{-1} \quad \checkmark$$

(3)

$$9.2.3 \quad F = Eq \quad \checkmark$$

$$F = (10\,000)(100 \times 10^{-6}) \quad \checkmark$$

$$F = 1 \text{ N} \quad \checkmark$$

(3)

[19]**QUESTION/VRAAG 10**

10.1 Maximum work done by the cell per coulomb of charge. ✓✓

Total energy transferred per coulomb of charge passing through the cell. ✓✓

*Maksimum arbeid deur die sel per coulomb lading verrig. ✓✓**Totale energy oorgedra per coulomb lading wat deur die sel beweeg. ✓✓*

(2)

$$10.2 \quad V = IR \quad \checkmark$$

$$4 = I(2) \quad \checkmark$$

$$I = 2 \text{ A} \quad \checkmark$$

(3)

$$10.3 \quad I_8 = 1 \text{ A} \quad \checkmark$$

Current in 8Ω is half of the current in the other branch*Stroom in 8Ω weerstand is helfte van stroom in die ander tak*

$$I_{\text{total}} = 2 + 1 \quad \checkmark = 3 \text{ A} \quad \checkmark$$

(3)

10.4 $V = I_{\text{total}} R_{\text{total}} \checkmark$

$$24 = (3)(R_{\text{total}})$$

$$R_{\text{total}} = 8 \, \Omega \checkmark$$

$$8 = R + R_{//}$$

$$\frac{1}{R_{//}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{//}} = \frac{1}{4} + \frac{1}{8}$$

$$R_{//} = 2,67 \, \Omega \checkmark$$

$$8 = R + 2,67 \checkmark$$

$$R = 5,33 \, \Omega \checkmark$$

OR/OF

$$V_{//} = 4 + 4 = 8V \checkmark$$

$$V_R = 24V - 8V = 16V \checkmark$$

$$R = \frac{V_R}{I} \checkmark$$

$$= \frac{16}{3} \checkmark$$

$$= 5,33 \, \Omega \checkmark$$

(5)

10.5 Increases/Neem toe $\checkmark$

Current through $2 \, \Omega$ resistor increases. $\checkmark \checkmark$

Stroom deur $2 \, \Omega$ resistor neem toe. $\checkmark \checkmark$

(3)

[16]

TOTAAL/TOTAL: 150